Techniques: PCR, Sequencing, RFLP

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Polymerase chain reaction (PCR)

Specifically targets and amplifies a SINGLE sequence from within a complex mixture of DNA.

- Invented by Dr. Kary Mullis in 1983

Takes advantage of basic requirements of replication

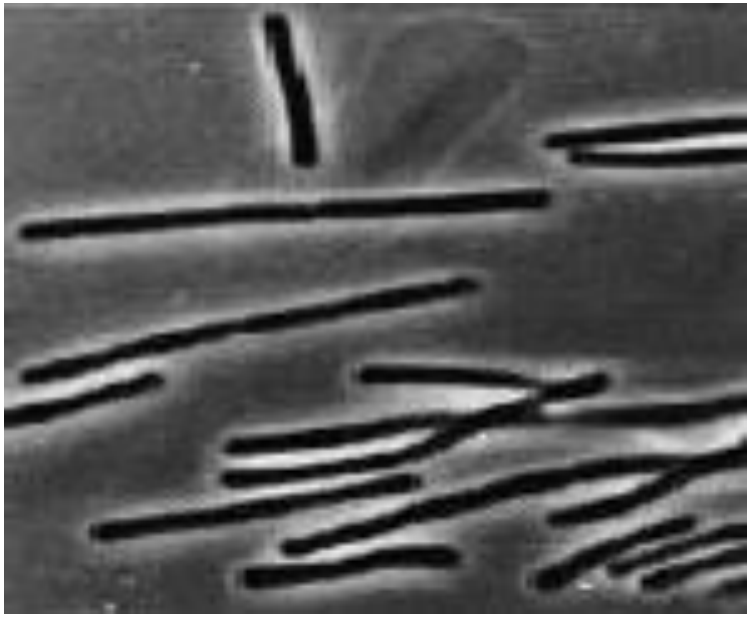
- A DNA template
- Nucleotides
- Primers
- polymerase

PCR is DNA replication of specific DNA fragment in vitro (in a test tube).

Taq DNA polymerase –Thermus aquaticus from hot springs purified since 1976.

Thermostable polymerases





Application of PCR:

- Diagnosis.
- Genetic counseling
- DNA profiling in forensic science.
- Paternity testing
- Recover DNA from rare sample archaeology.

Templates for PCR

- Dried blood
- Semen stains
- Vaginal swabs
- Single hair
- Fingernail scrapings
- Egyptian mummies
- Toothbrushes

- Basic reaction
- Oligonucleotide primers
- Design to flank the desired sequence:two oligonucleotide primers complementary to opposite strands of a duplex DNA
- Steps include: (20-40 repeats)
    - ** Denaturation at 94°C
    - ** Primer annealing at Tm-5°C
    - ** Extension at 72°C
- Mg+ content – 1-4mM (stabilize duplex)

Thermocycler, PCR machine:

- Automatically change temperature
- Temperature gradient
- The DNA, Taq polymerase, buffer, nucleoside triphosphate, primers placed in a thin walled tube placed in the machine.







Amplification

- Able to amplify from very small amount of DNA.
- If your DNA is subjected to amplification 5 cycles you will get $2_5 = 64$ copies of DNA
- High sensitivity
- Specificity – Primers design.

Problem with PCR

- Contamination of sample or reagents, theoretically one molecule can amplify.

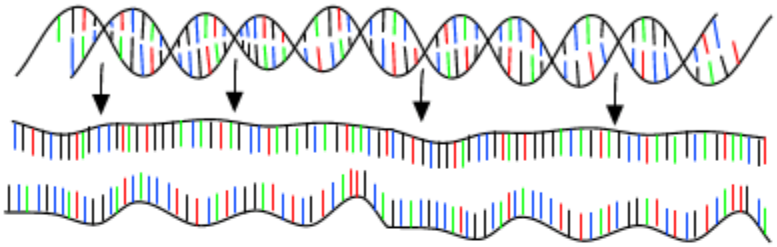
Primers

- ►Designing primers

  17-30 nucleotides long usually about 20bp
- A GC content about 50% - control melting temperature > 54°C
- Have no or minimum secondary structure (hairpin)
- No complementary between the two primers (primer dimer)

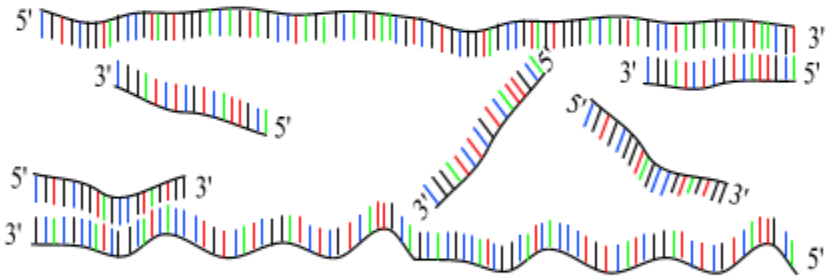
PCR : Polymerase Chain Reaction

30 - 40 cycles of 3 steps :



Step 1 : denaturation

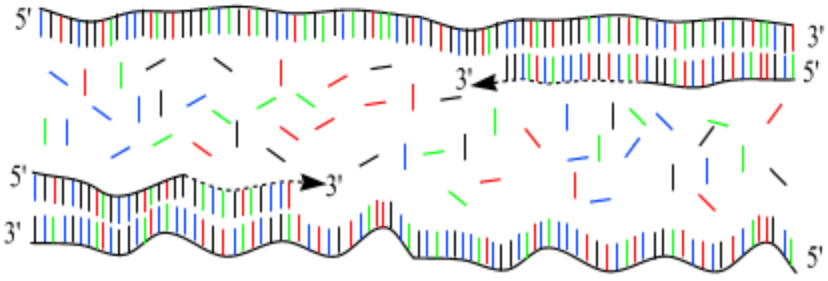
1 minut 94 °C



Step 2 : annealing

45 seconds 54 °C

forward and reverse primers !!!

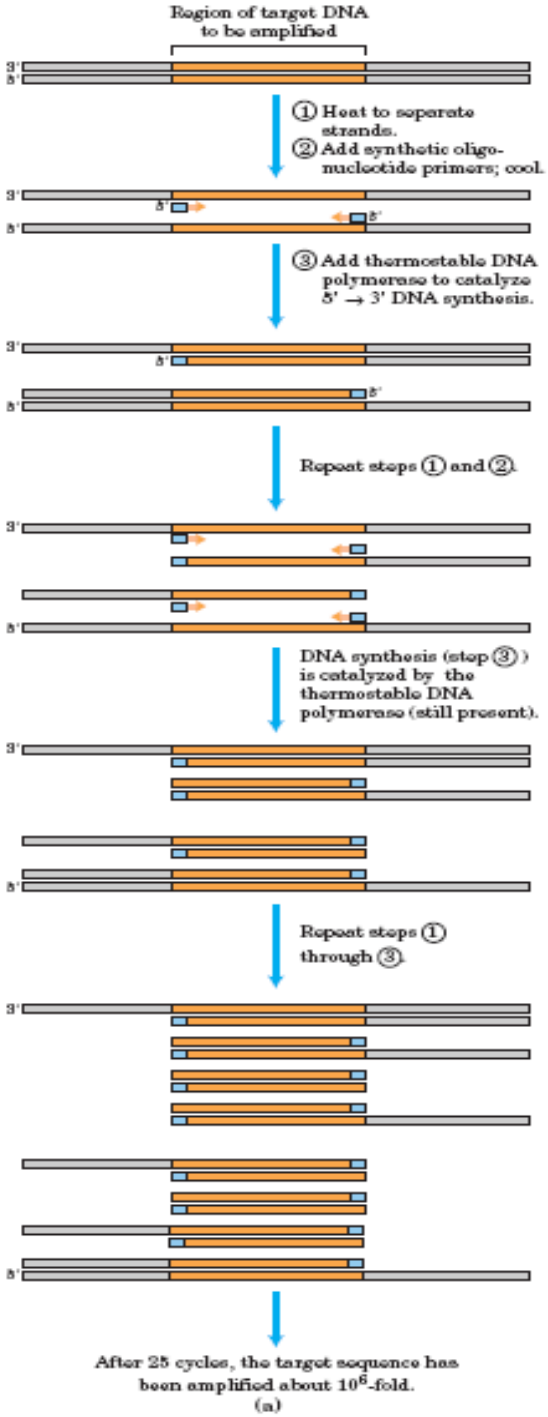


Step 3 : extension

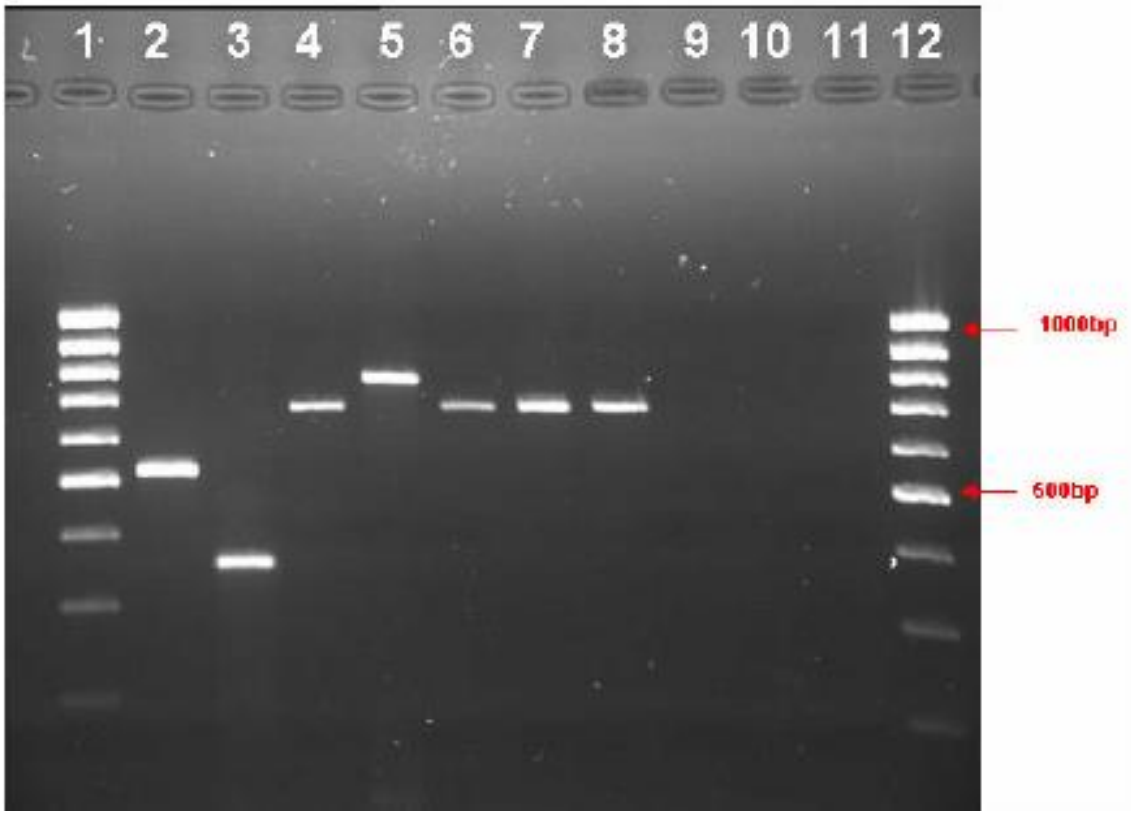
2 minutes 72 °C

only dNTP's

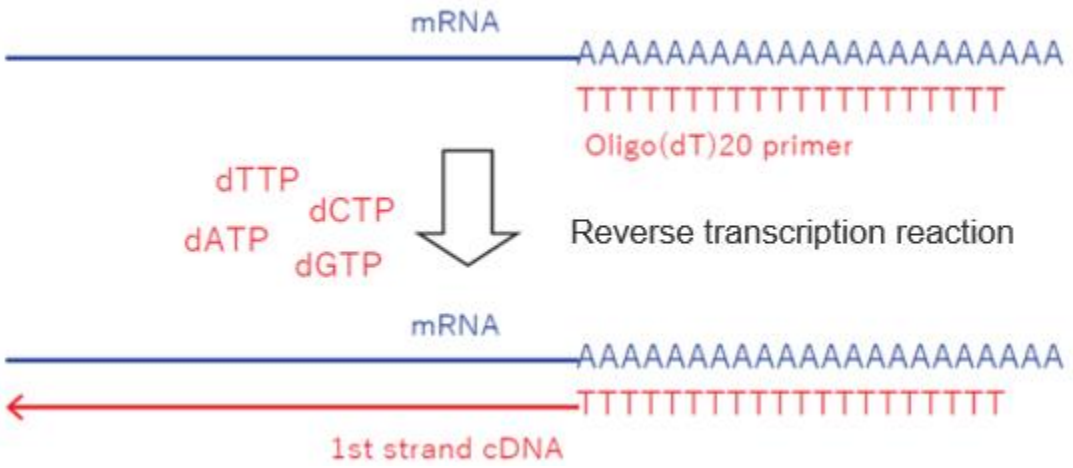
(Andy Vierstraete 1999)



PCR product



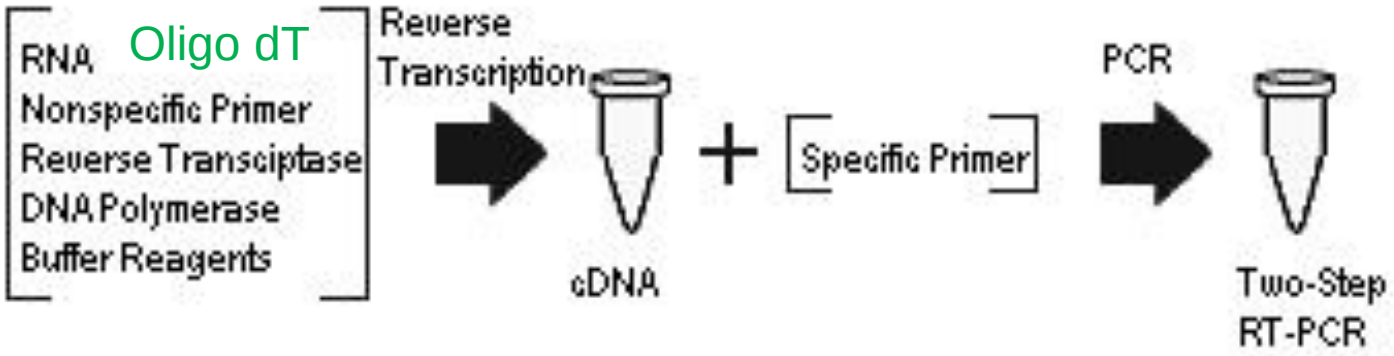
RT-PCR: Reverse transcription polymerase chain reaction



PCR

RT- PCR: Reverse transcription polymerase chain reaction

1- RT- step:



2- PCR step:

DNA sequencing

Sanger's method,

Dideoxy sequencing

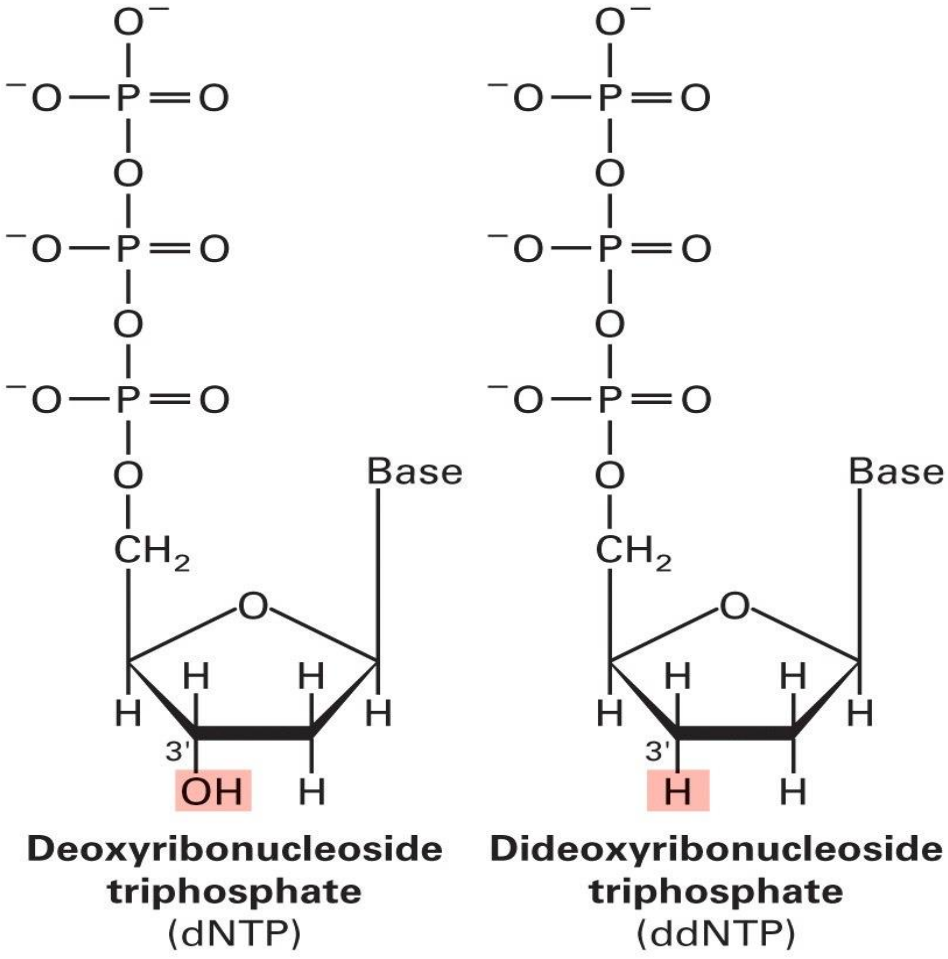
or Chain termination

Four steps:

1. Denaturation
2. Primer annealing and extension
3. Termination (ddNTP)
4. Gel electrophoresis

Dideoxise block elongation

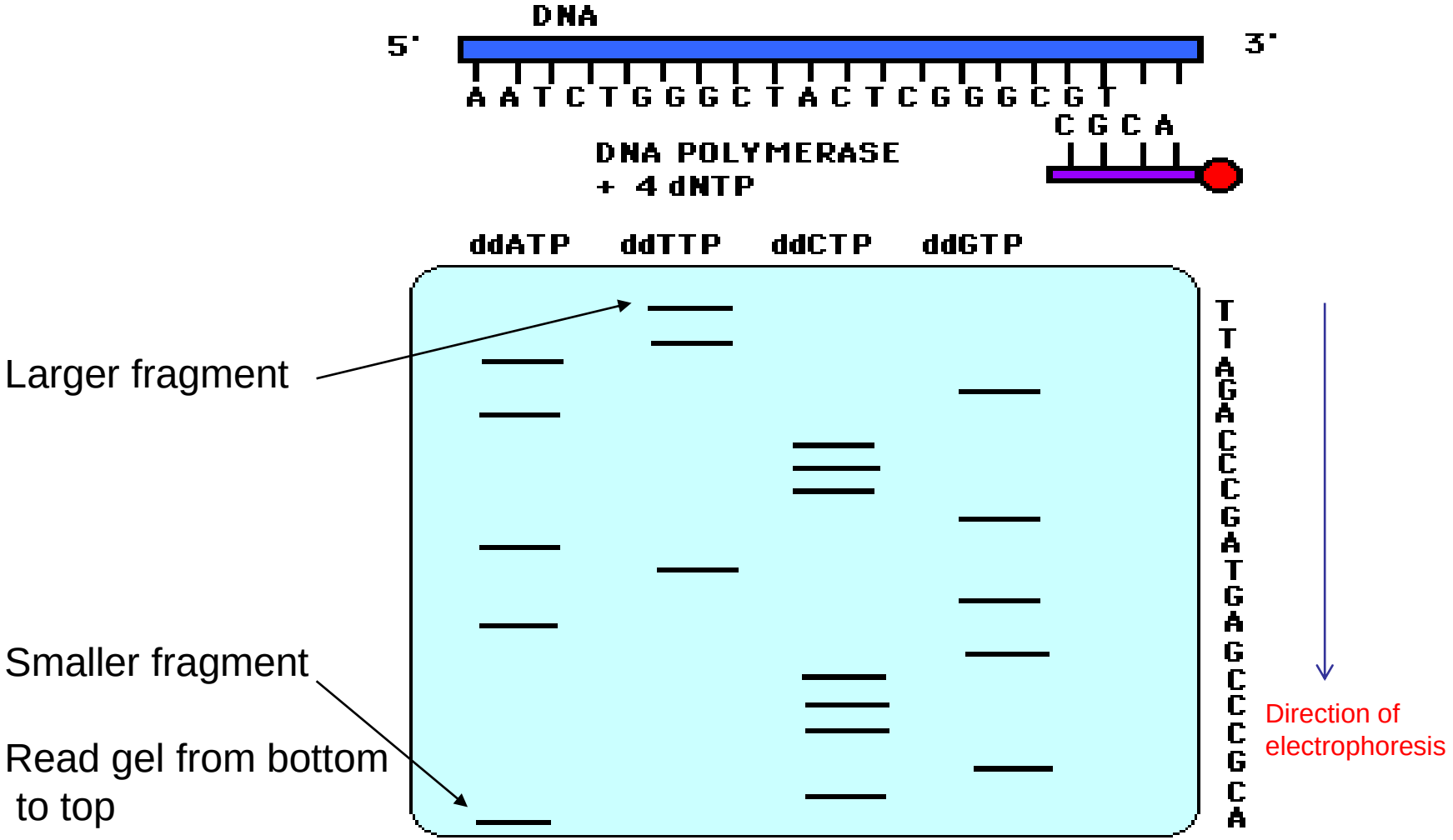
ddNTP modified nucleotides, when integrated into a sequence, prevent addition of further nucleotides. Since phosphodiester bond cant form bw. dideoxynucleotide and the next incoming nucleotide, and thus the DNA **chain is terminated**.

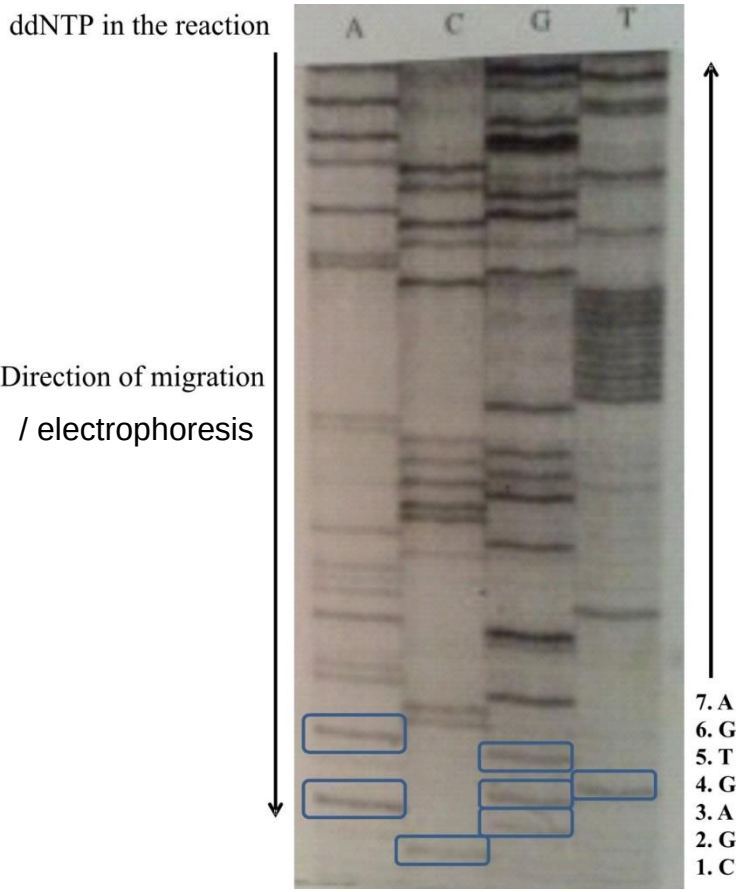


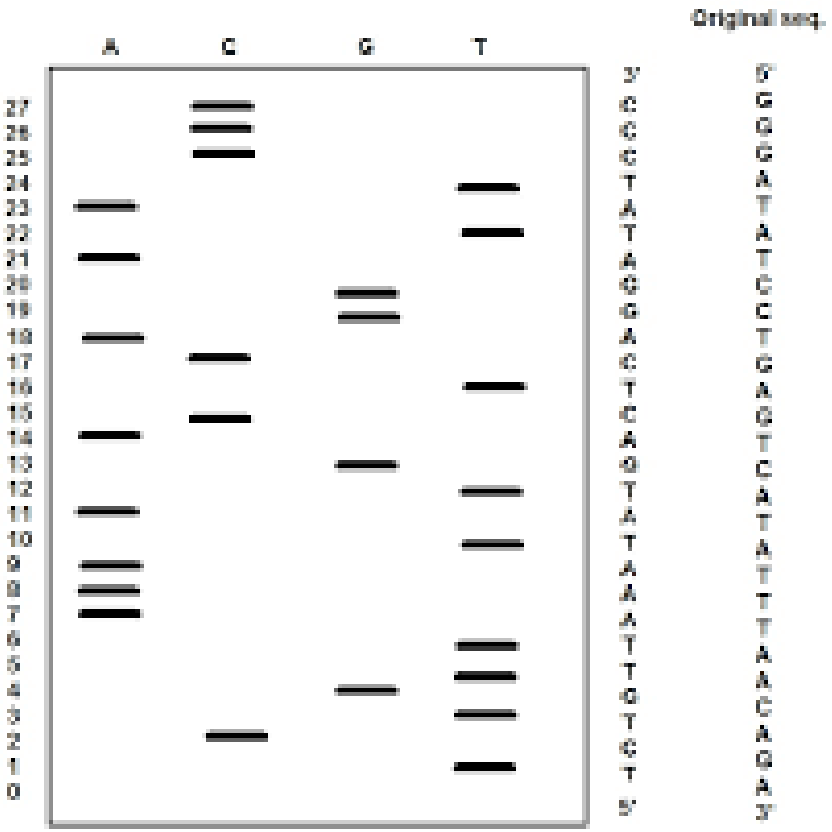
In both cases, DNA polymerase will incorporate nucleoside monophosphates, but…..

3’ OH can be used for phosphodiester bond	No 3’ OH: DNA synthesis terminates

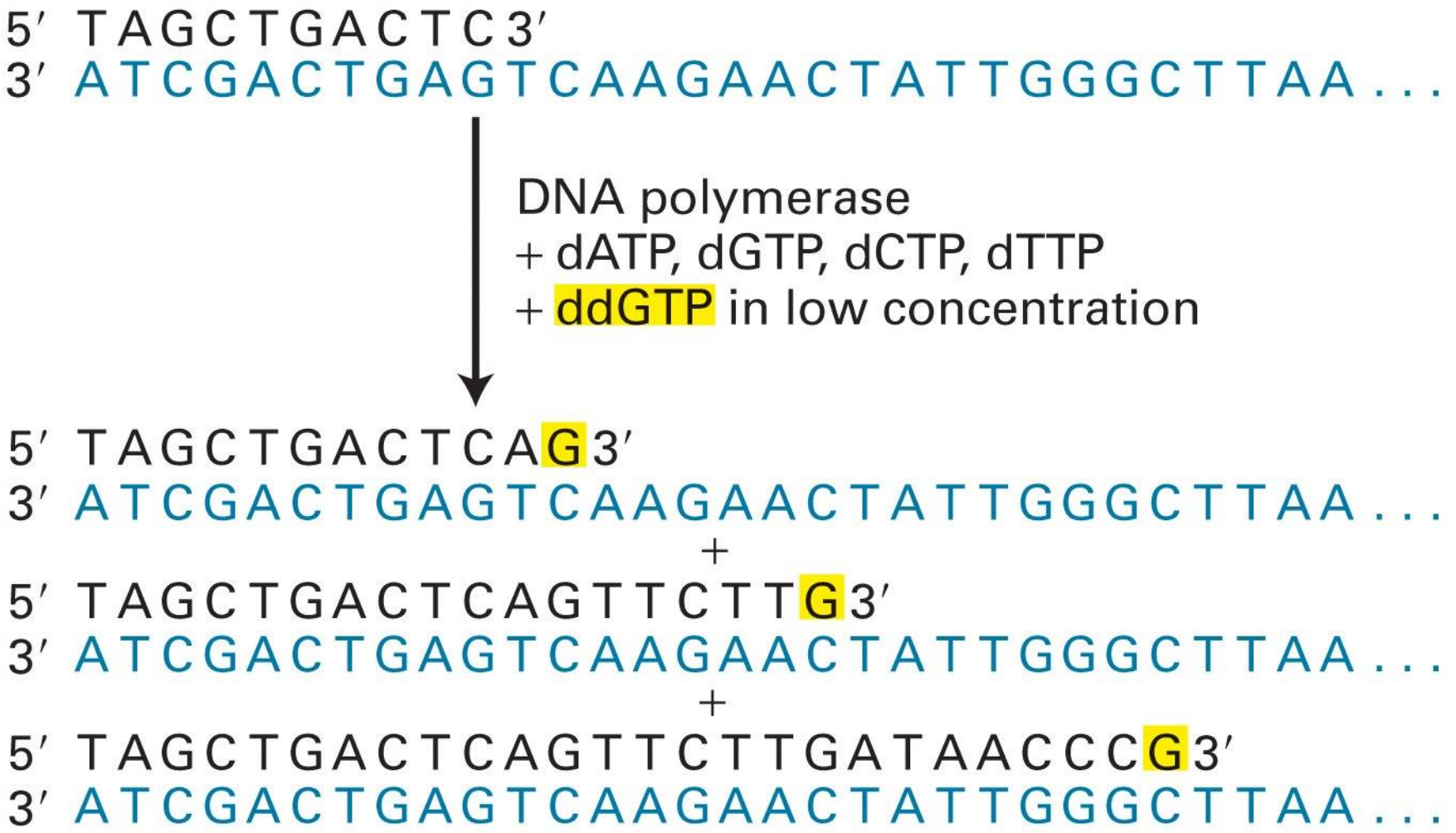
- **G" tube:** all four dNTP's, ddGTP and DNA pol
- **"A" tube:** all four dNTP's, ddATP and DNA pol
- **"T" tube:** all four dNTP's, ddTTP and DNA pol
- **"C" tube:** all four dNTP's, ddCTP and DNA pol

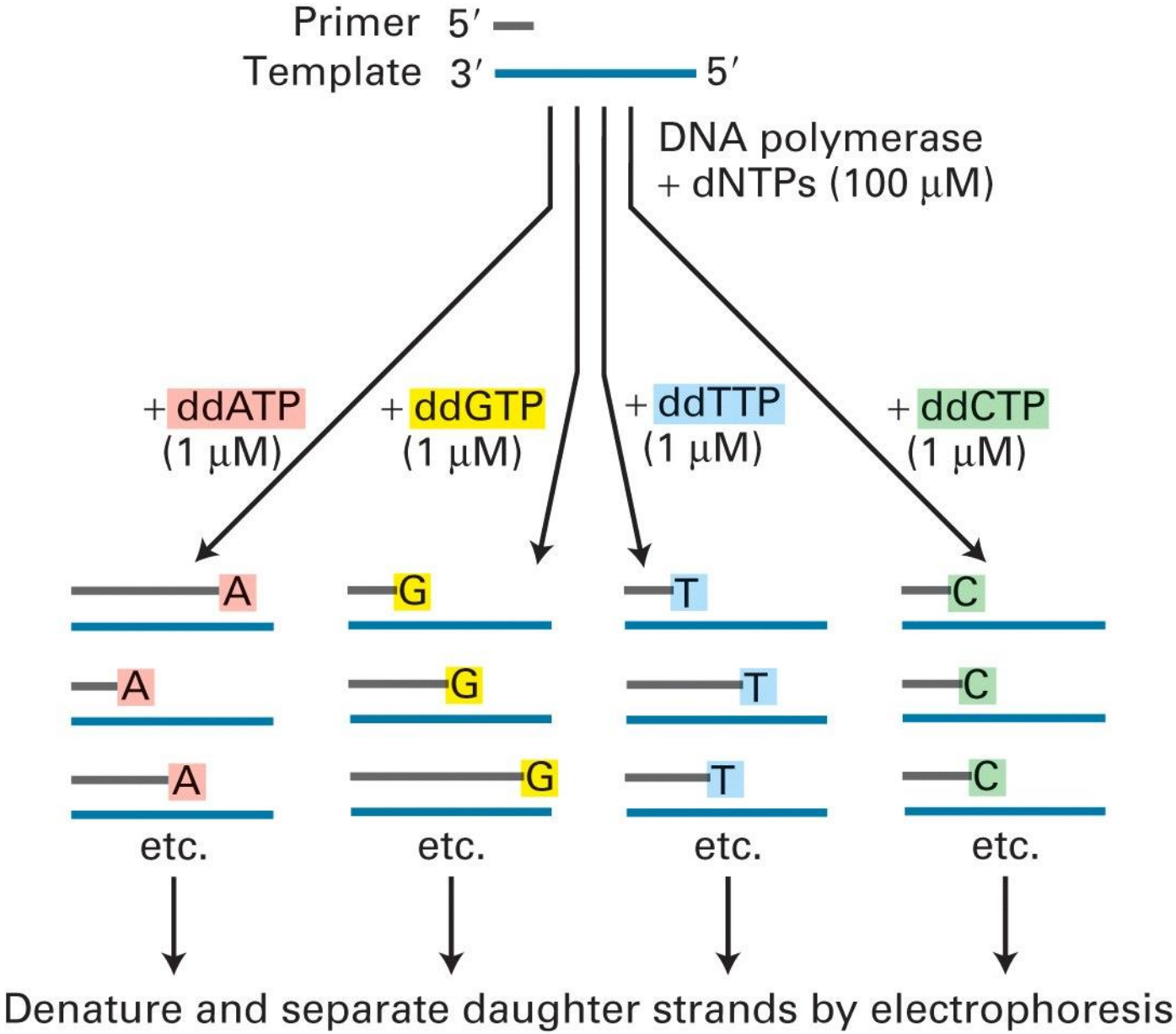




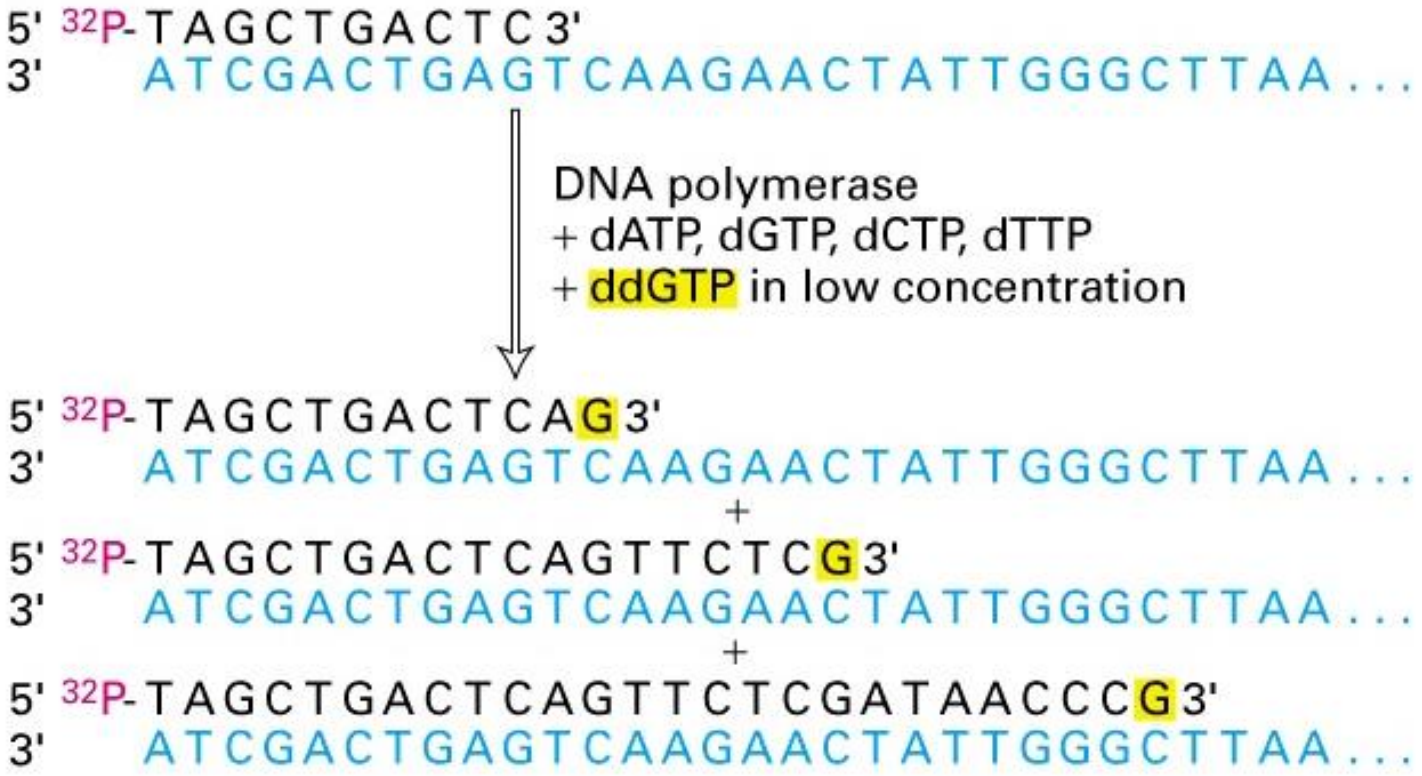


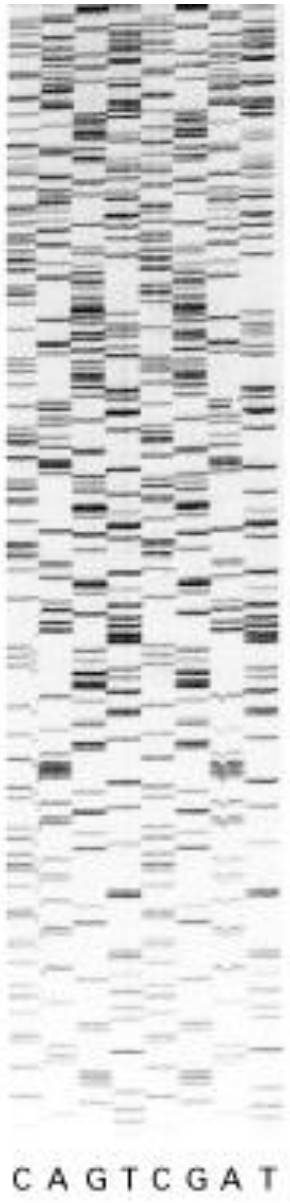
From bottom to top = sequence of synthesized DNA
From top to bottom = sequence of template DNA


5′ TAGCTGACTC 3′
3′ ATCGACTGAGTCAAGAACTATTGGGCTTAA . . .
DNA polymerase
+ dATP, dGTP, dCTP, dTTP
+ ddGTP in low concentration
5′ TAGCTGACTCAG 3′
3′ ATCGACTGAGTCAAGAACTATTGGGCTTAA . . .
+
5′ TAGCTGACTCAGTTCTTG 3′
3′ ATCGACTGAGTCAAGAACTATTGGGCTTAA . . .
+
5′ TAGCTGACTCAGTTCTTGATAACCCG 3′
3′ ATCGACTGAGTCAAGAACTATTGGGCTTAA . . .

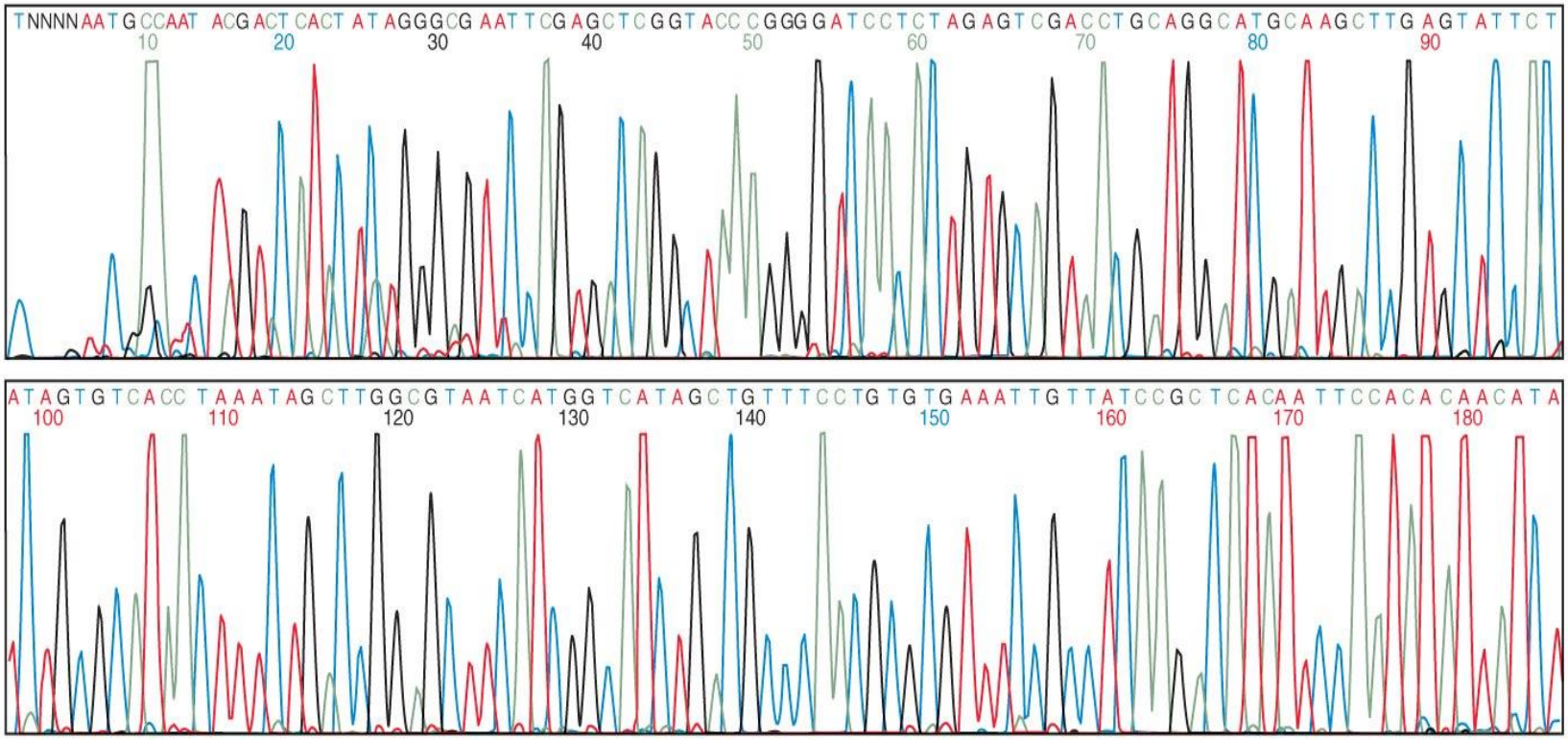

Primer 5′
Template 3′
5′
DNA polymerase
+ dNTPs (100 μM)
+ ddATP
(1 μM)
+ ddGTP
(1 μM)
+ ddTTP
(1 μM)
+ ddCTP
(1 μM)
A
A
A
G
G
G
T
T
T
C
C
C
etc.
etc.
etc.
etc.
Denature and separate daughter strands by electrophoresis

DNA sequencing: the Sanger (dideoxy) method





Automated DNA sequencing involves use of four different fluorescent primers allowing the simultaneous detection of all four reactions in one sample.



restriction fragment length polymorphism, or **RFLP**,

the DNA sample is broken into pieces by restriction enzymes → the resulting *restriction fragments* are separated according to their lengths by gel electrophoresis.

Restriction fragment length polymorphism (RFLP)

a technique invented in 1984 by the English scientist Alec Jeffreys during research into hereditary diseases.

Genetic polymorphism: the inherited genetic differences among individuals in over 1% of normal population.

RFLP technique exploits these differences in DNA sequences to recognize & study both intraspecies and interspecies variation.

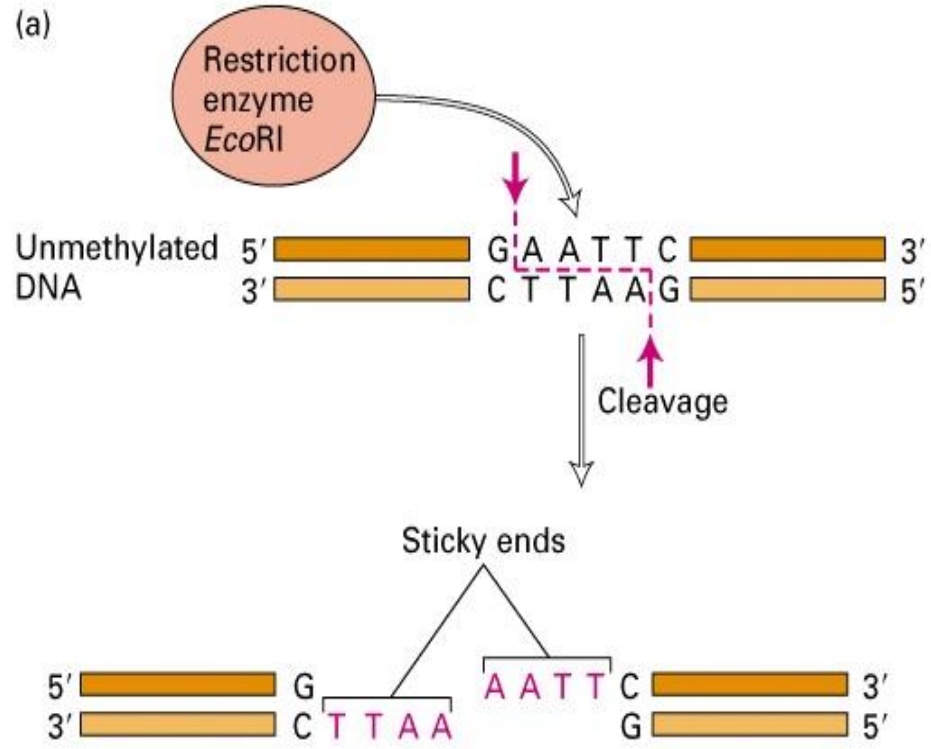
Restriction enzymes

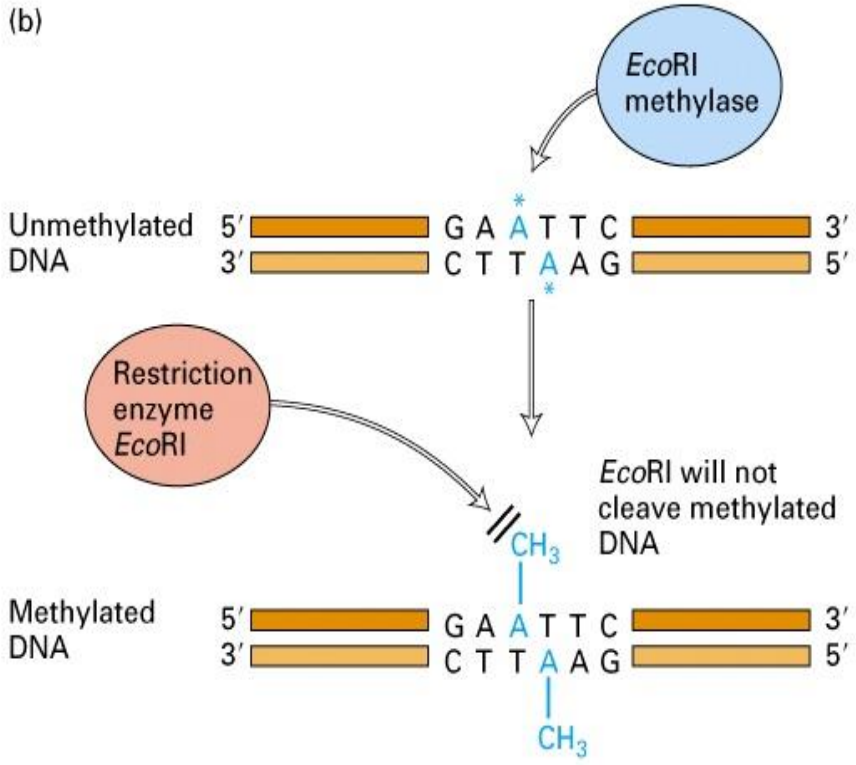
DNA-cutting enzymes found in bacteria and harvested from them for use. Since they cut within the molecule = **restriction endonucleases**.

-Different species of bacteria make different restriction nucleases, which protect them from viruses by degrading foreign incoming viral DNA.

-These sequences, where they occur in the genome of the bacterium itself, are protected from cleavage by methylation at A / C residue.

Restriction enzymes:
cut DNA molecules at specific sequences

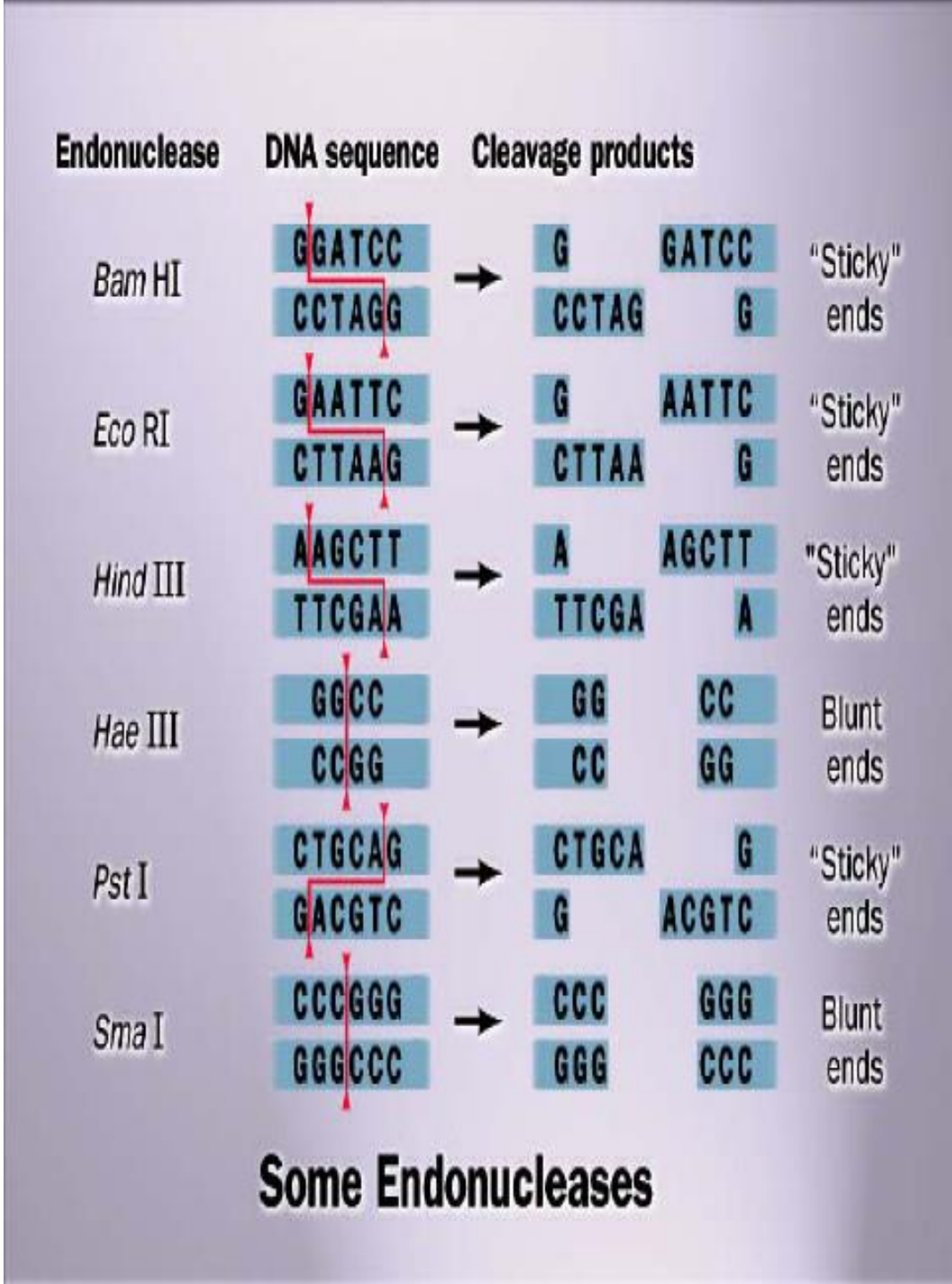




Acts on palindrome sequences (have symmetry) :

Cut both strands of DNA.

Yielding either blunt end or sticky ends.



Sickle-cell Anemia

caused by a mutation in the **beta-globin** gene.

difference between normal $\boldsymbol{B^A}$ allele & sickle-cell $\boldsymbol{B^S}$ allele is a single-nucleotide substitution (**A→T**)

The sequence of the $\boldsymbol{B^A}$ allele (**CCTGAGG**)

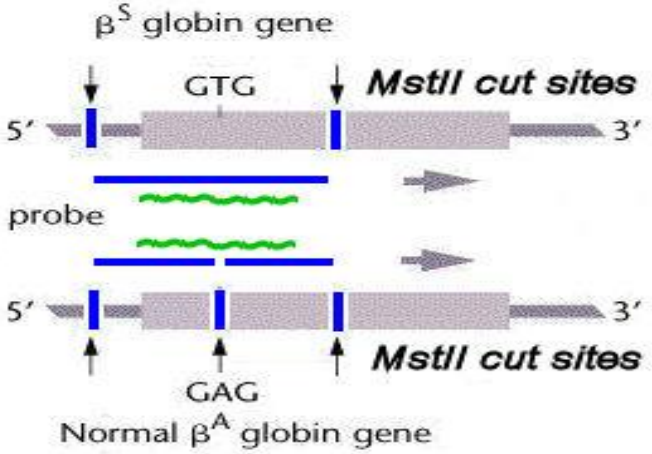
The sequence of the $\boldsymbol{B^S}$ allele (**CCTG*T*GG**).

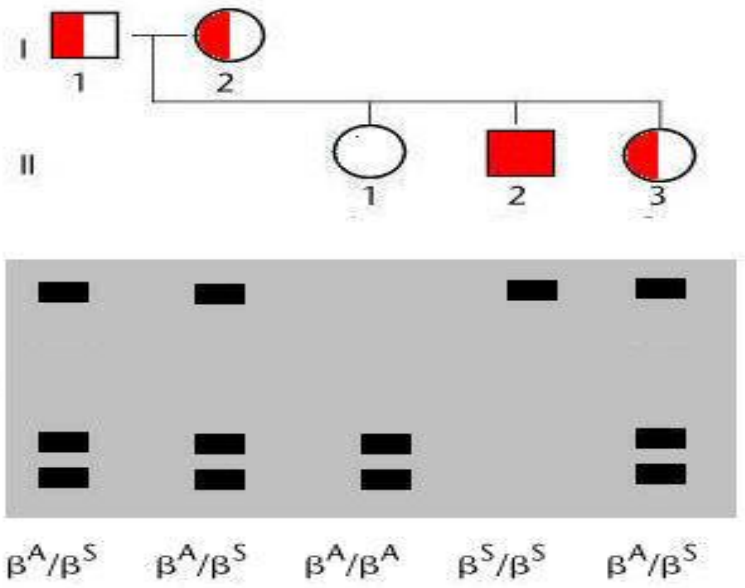
***MstII* restriction site** (**CCTNAGG**)→ cuts the $\boldsymbol{B^A}$ allele

Protocol:

total **DNA** →digested with ***MstII*** →run in electrophoresis → The blot is hybridized with a **probe** specific for the **beta-globin** gene (**Southern Blot)**.

RFLP / *MstII* test for Sickle-Cell Anemia





standard ***B*^A** allele→probe sticks to 2 small ***MstII*** fragments → produces **two smaller bands**

sickle-cell ***B*^S** allele →probe sticks to the single large fragment →**one larger band**.

AA homozygote shows the two-band pattern,
SS homozygote (with **sickle-cell anemia**) shows the one-band pattern,
AS heterozygote (with **sickle-cell *trait***) shows **all three bands**.